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FACULTY OF COMPUTING

Department of Software Engineering



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Operating System and System Programming (Seng2034)



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Introduction

Operating systems are the core foundation of any computing system. They act as a bridge between the user and the computer hardware, managing important resources such as memory, processor usage, and storage while ensuring that programs run smoothly and securely.

In recent years, Linux-based operating systems have become increasingly popular because of their flexibility, strong security, and open-source nature. One of the most widely used distributions is Ubuntu 24.04.1, which is well known for its stability, ease of use, and long-term support (LTS).

Ubuntu is commonly used in educational institutions, software development environments, and even in enterprise systems. Its reliability and strong community support make it an ideal platform for learning and practical implementation. In the field of system programming, Ubuntu is especially important because it allows direct interaction with low-level system calls such as `getuid()`, which is used to identify the user running a particular process.

The main motivation behind this project is to gain hands-on experience by installing Ubuntu in a virtual environment and exploring how system-level operations work. It also aims to provide a better understanding of how Linux manages user identities and how system calls like `getuid()` function in real scenarios.

Using virtualization tools makes this process safe and flexible, as it allows the operating system to run without affecting the main system installed on the computer.

1. Objectives

The main aim of this project is to develop both practical and theoretical understanding of operating systems through hands-on experience. The specific objectives are:

- To install Ubuntu 24.04.1 using virtualization software and become familiar with the installation process
- To understand each step involved in setting up a Linux-based operating system
- To explore the functionality and importance of the `getuid()` system call in identifying users within the system
- To study the different file systems supported by Ubuntu and their purposes
- To identify common problems that may occur during installation and learn how to solve them effectively
- To evaluate the advantages and limitations of using Ubuntu as an operating system
- To understand the concept of virtualization and its importance in modern computing environments

2. Requirements

i. Hardware Requirements

To run Ubuntu efficiently in a virtual environment, the system should meet the following requirements:

- A processor that supports virtualization (Intel VT-x or AMD-V)
- At least 4 GB of RAM (8 GB is recommended for better performance)
- Around 30–40 GB of free disk space
- A stable internet connection for downloading and updates

ii. Software Requirements

- Virtualization software such as Oracle VM VirtualBox or VMware Workstation
- Ubuntu 24.04.1 ISO file
- A host operating system (Windows, Linux, or macOS)

3. Installation Steps

Step 1: Download Ubuntu ISO

Go to the official Ubuntu website and download the Ubuntu 24.04.1 ISO file. Make sure you select the correct 64-bit version and wait until the download is fully completed before proceeding.

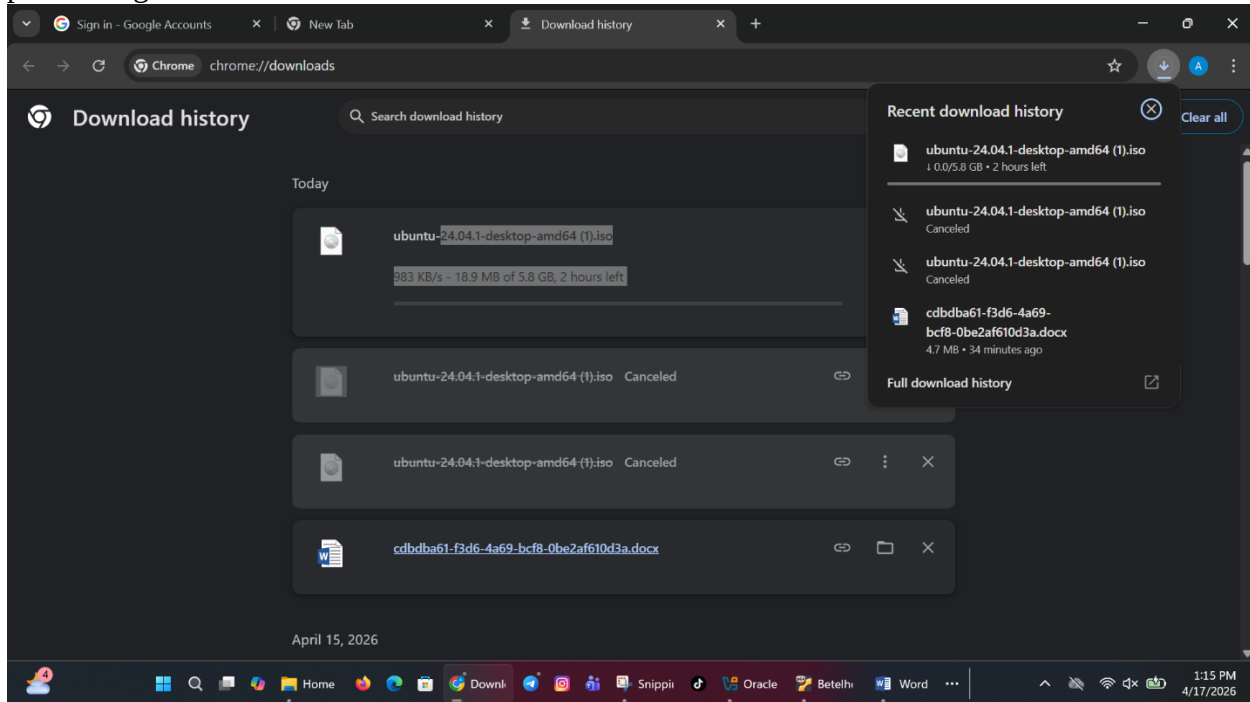


Figure 1: Download Ubuntu ISO file

Step 2: Install VirtualBox

Run the VirtualBox setup file that you downloaded. Follow the installation wizard by clicking “Next” on each step, and keep the default settings unless otherwise required. Finish the installation and open VirtualBox.

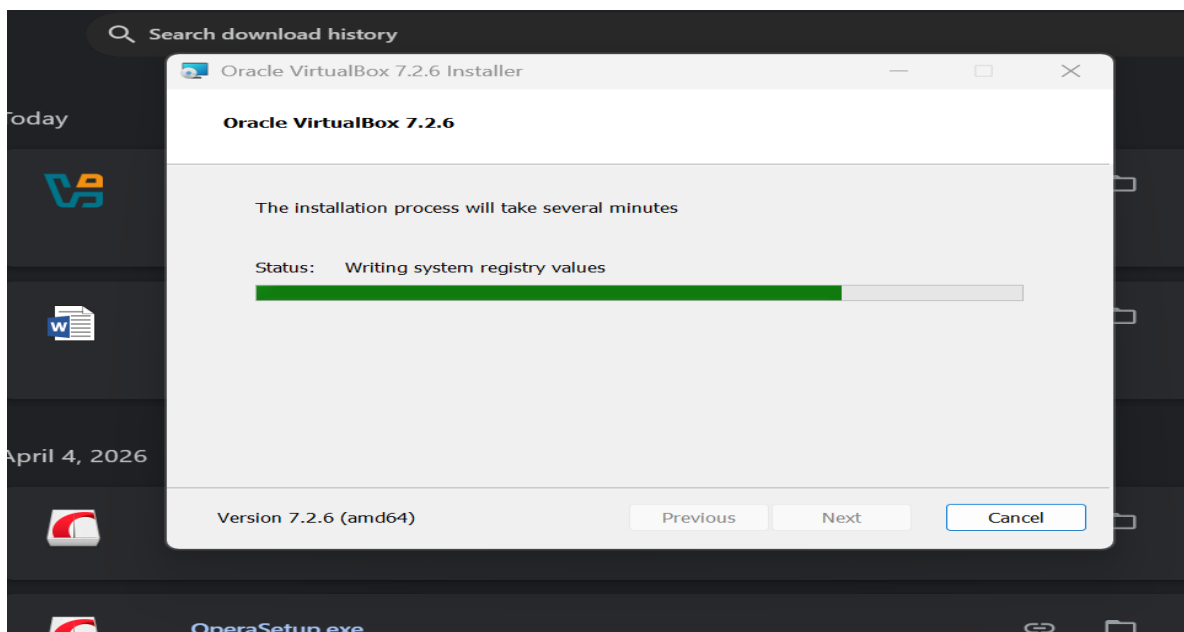


Figure 2: VirtualBox installation setup

Step 3: Create a New Virtual Machine

Open VirtualBox and click “New” to create a virtual machine. Enter a name for the VM, choose “Linux” as the type, and select “Ubuntu (64-bit)” as the version. Then proceed to the next step.

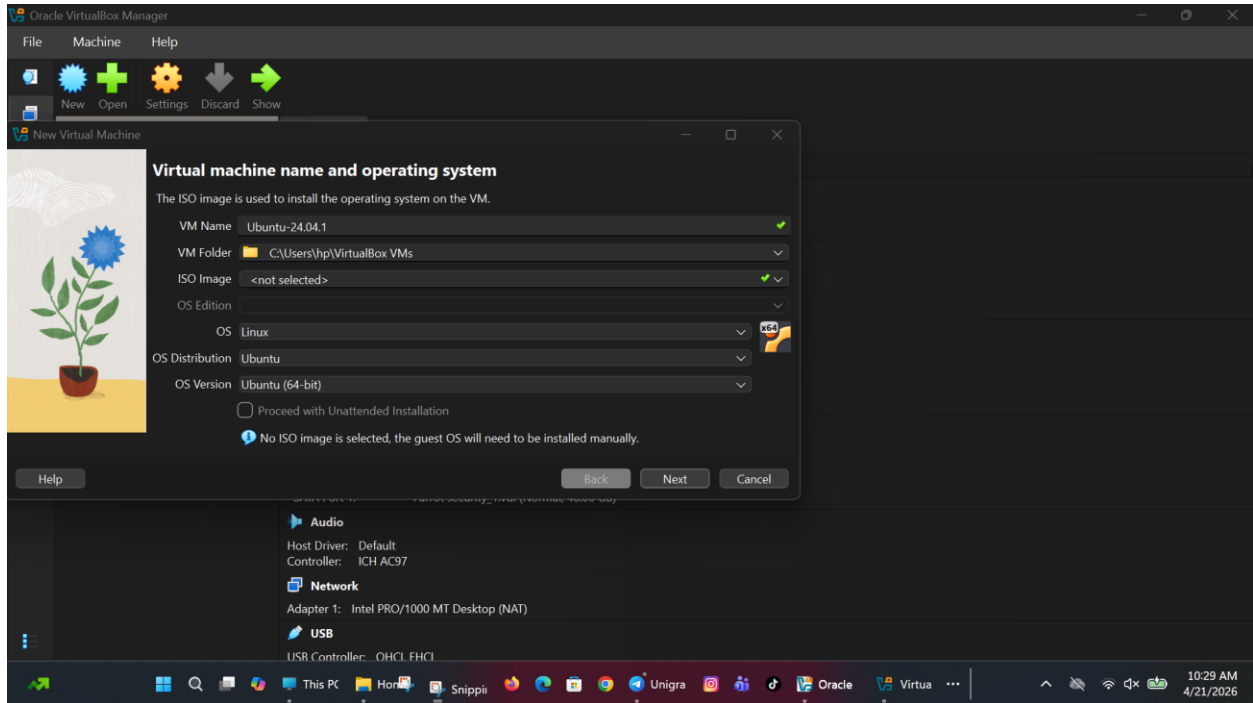


Figure 3: Creating new virtual machine

Step 4: Allocate RAM

Assign memory (RAM) to your virtual machine. It is recommended to allocate at least 2048 MB (2 GB), but for better performance, select 4096 MB (4 GB) or more depending on your system capacity. Click “Next” to continue.

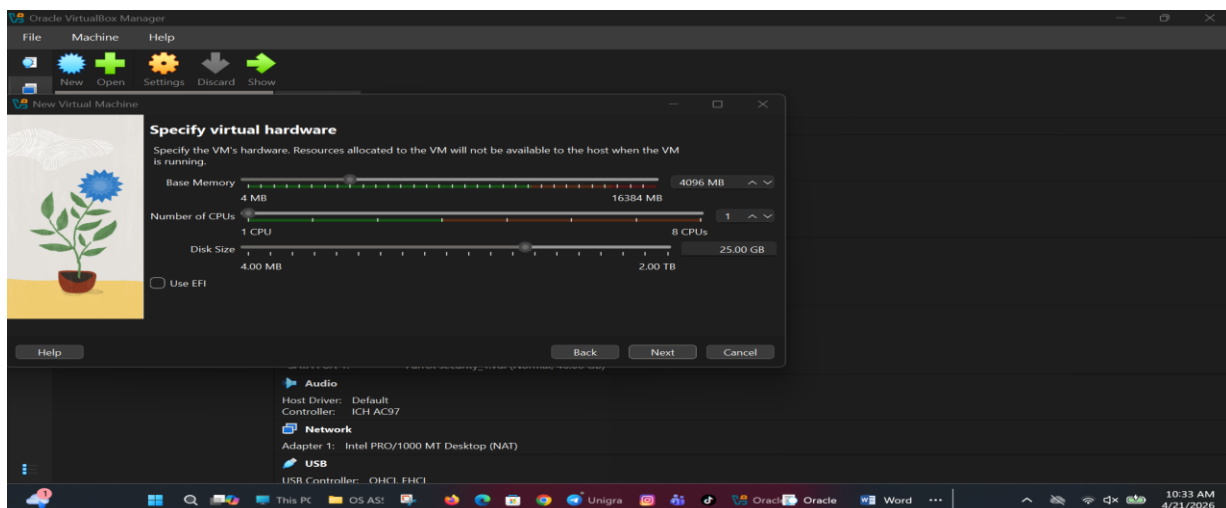


Figure 4: Allocating RAM to VM

Step 5: Create Virtual Hard Disk

Select “Create a virtual hard disk now” and continue. Choose the VDI (VirtualBox Disk Image) format, set it to dynamically allocated storage, and assign a size of around 40 GB to ensure enough space for Ubuntu and applications..

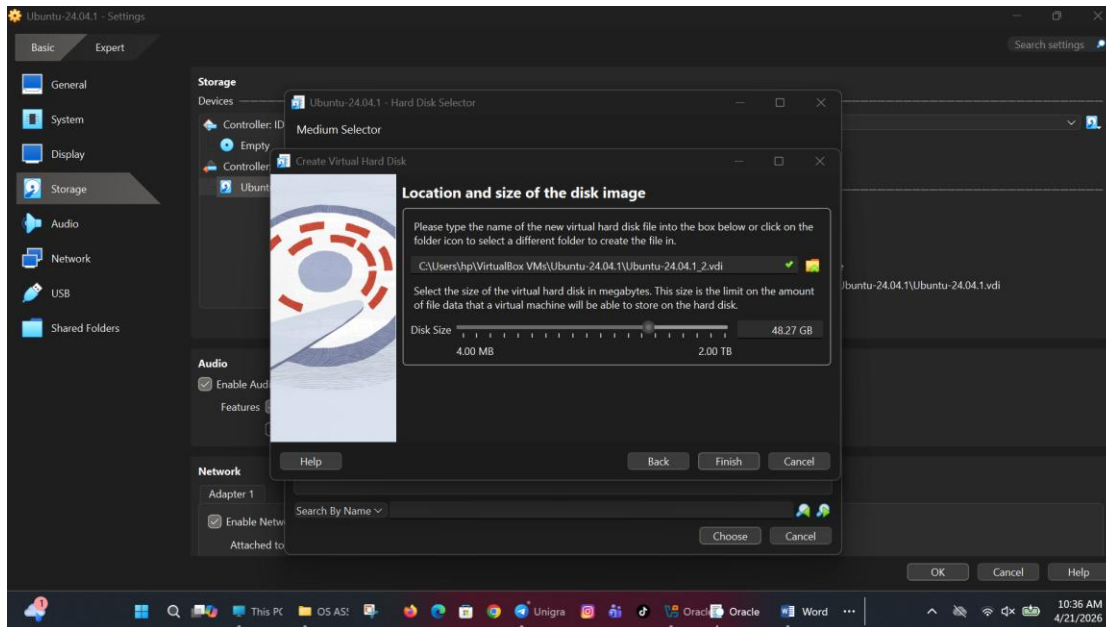


Figure 5: Creating virtual hard disk

Step 6: Attach ISO File

Go to Settings → Storage, select “Empty” under the controller section, then click the disk icon next to Optical Drive. Choose “Choose a disk file” and locate the downloaded Ubuntu ISO file.

Ensure the ISO is properly attached and visible under the controller as a virtual optical drive.

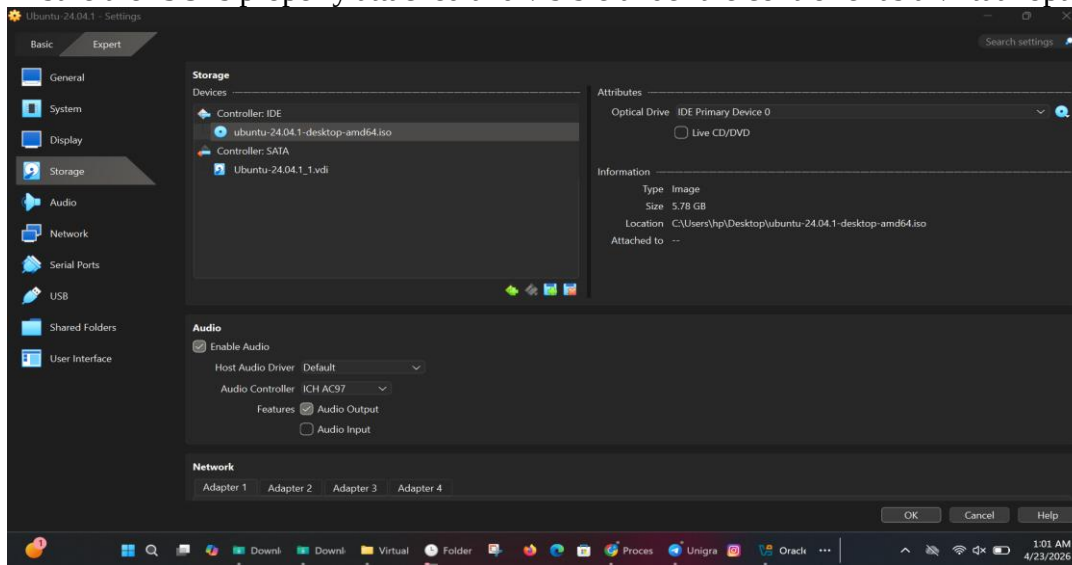


Figure 6: Attaching Ubuntu ISO file

Step 7: Start Installation

Click **Start** to run the virtual machine. When the Ubuntu screen appears, select **“Install Ubuntu”** to begin the installation process.

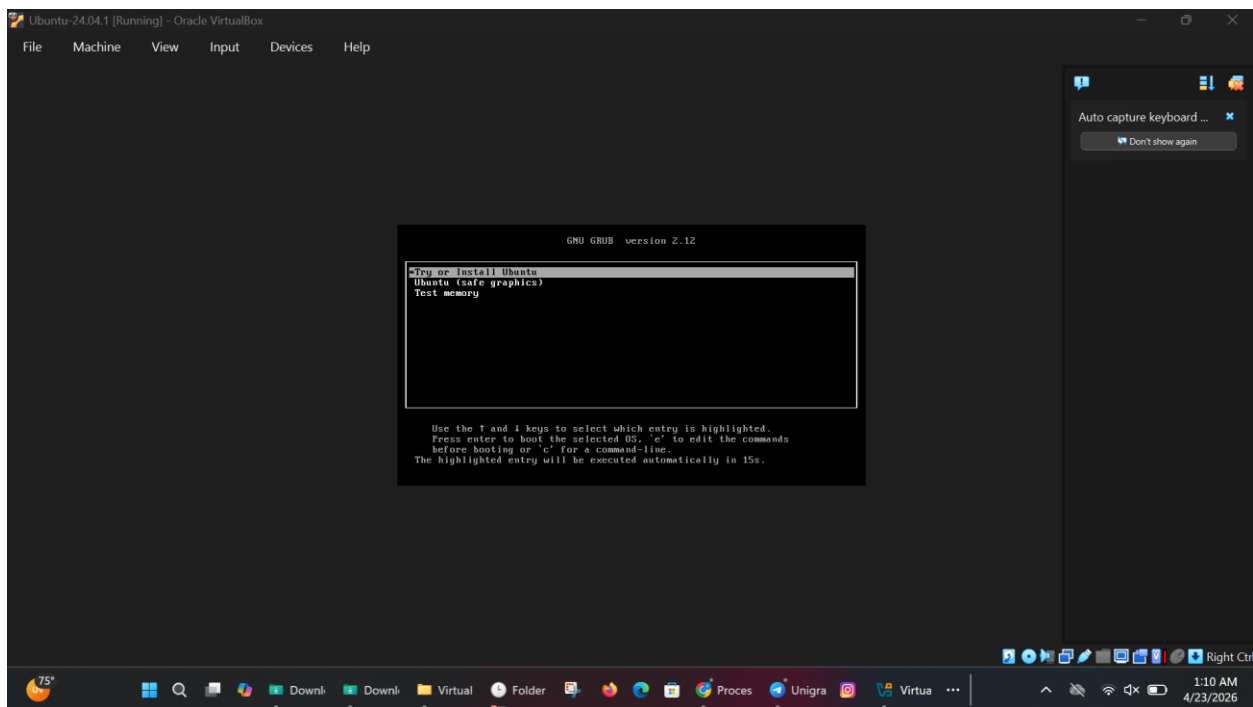


Figure 7: Starting Ubuntu installation

Step 8: Select Language and Keyboard

Choose your preferred installation language from the list. Then select your keyboard layout based on your region or preference and continue to the next step.

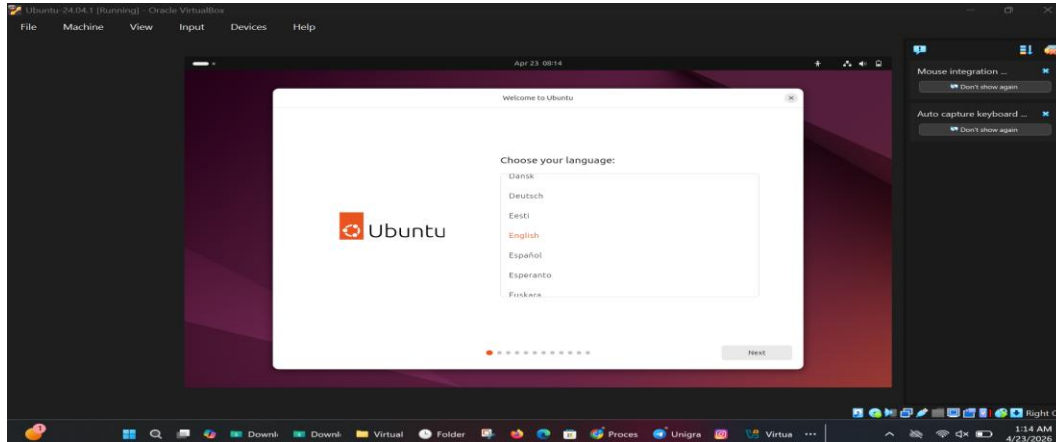


Figure 8: Selecting language and keyboard

Step 9: Installation Type

Select “Normal Installation” and proceed.

Step 10: Disk Partitioning

Choose “Erase disk and install Ubuntu” to allow Ubuntu to use the virtual hard disk you created. Confirm the selection and continue, as this only affects the virtual machine, not your real computer.

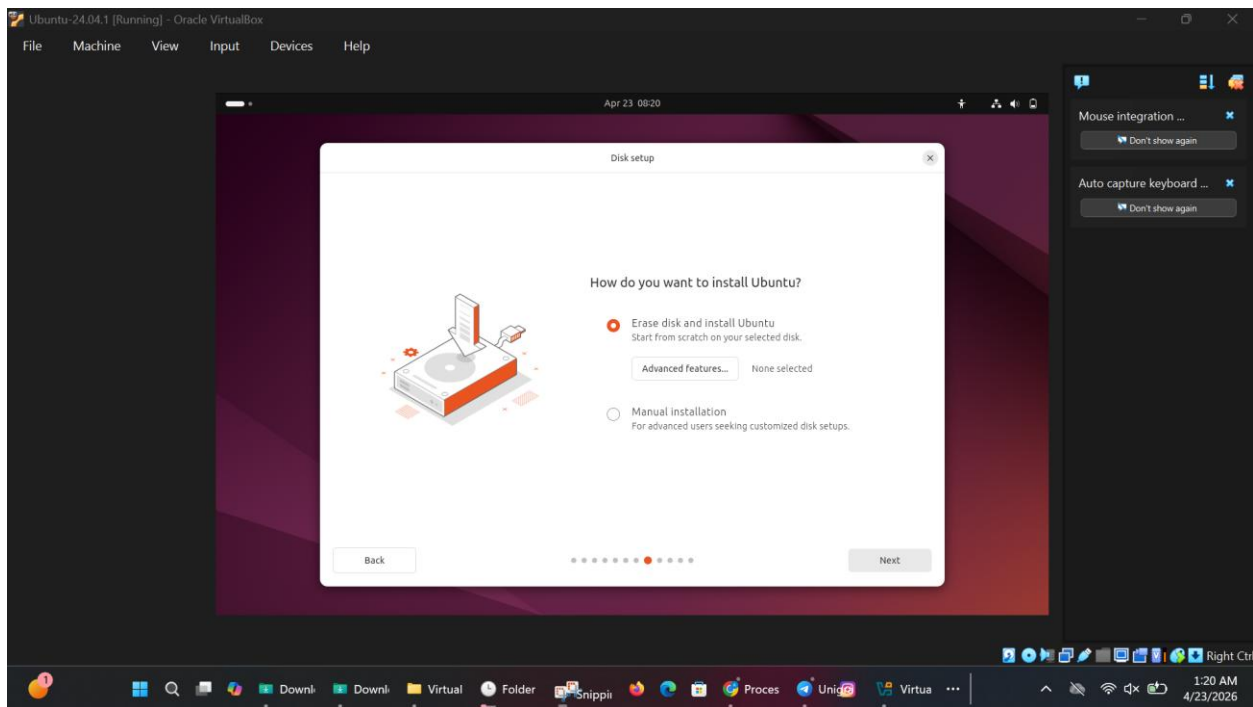


Figure 10: Disk partitioning setup

Step 11: Create User Account

Enter your full name, computer name, username, and password. Make sure the password is strong and memorable, as it will be used to log in to Ubuntu..

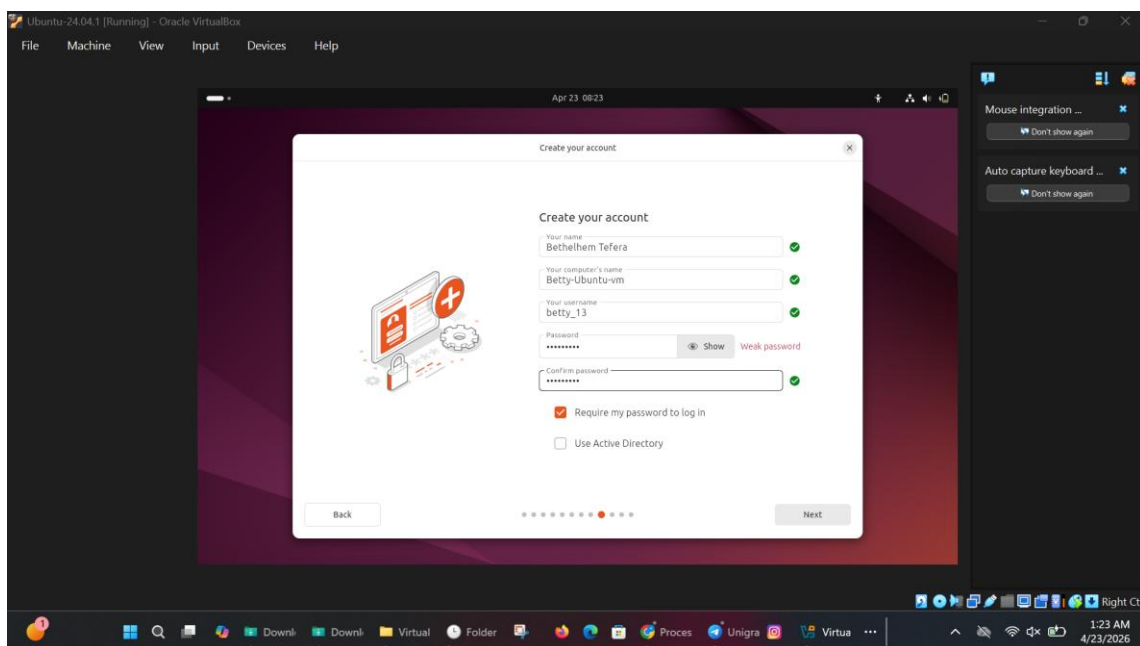


Figure 11: Creating user account

Step 12: Installation Process

The system will now begin copying files and installing Ubuntu. This process may take several minutes depending on your system speed. Wait patiently until it completes.

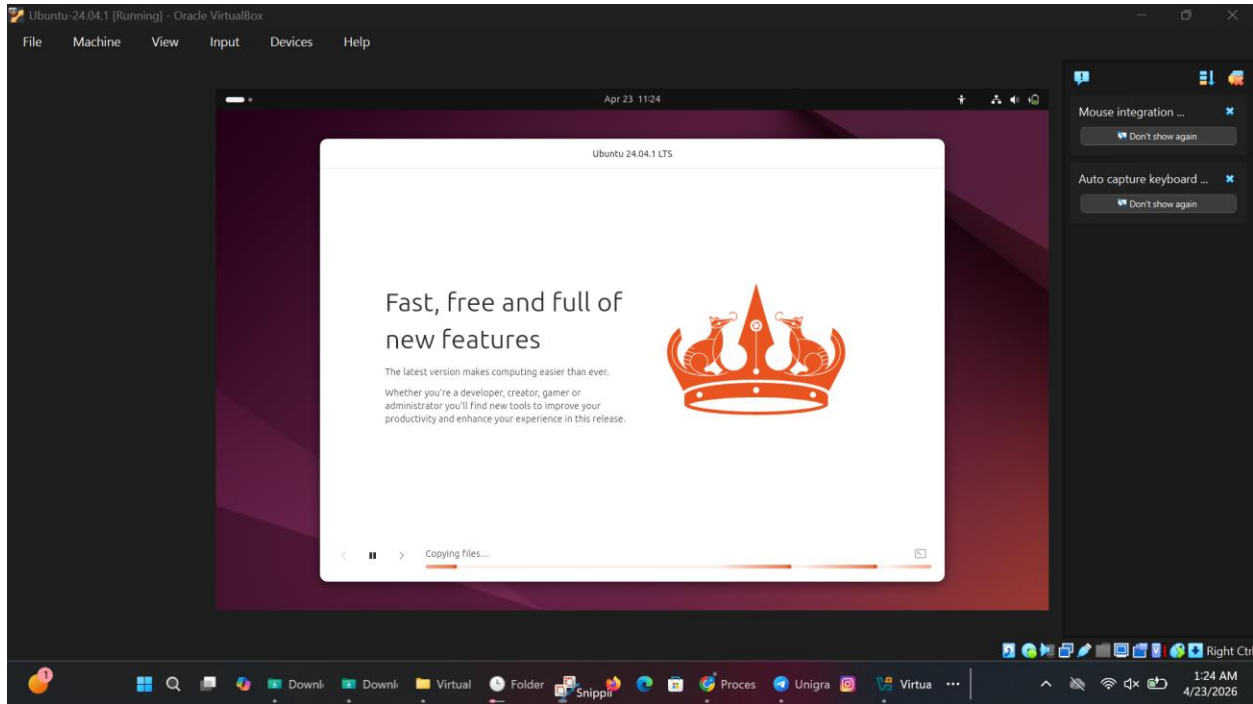


Figure 12: Ubuntu installation process

Step 13: Restart System

Once installation is complete, click “Restart Now” when prompted. The virtual machine will reboot to complete the setup process.

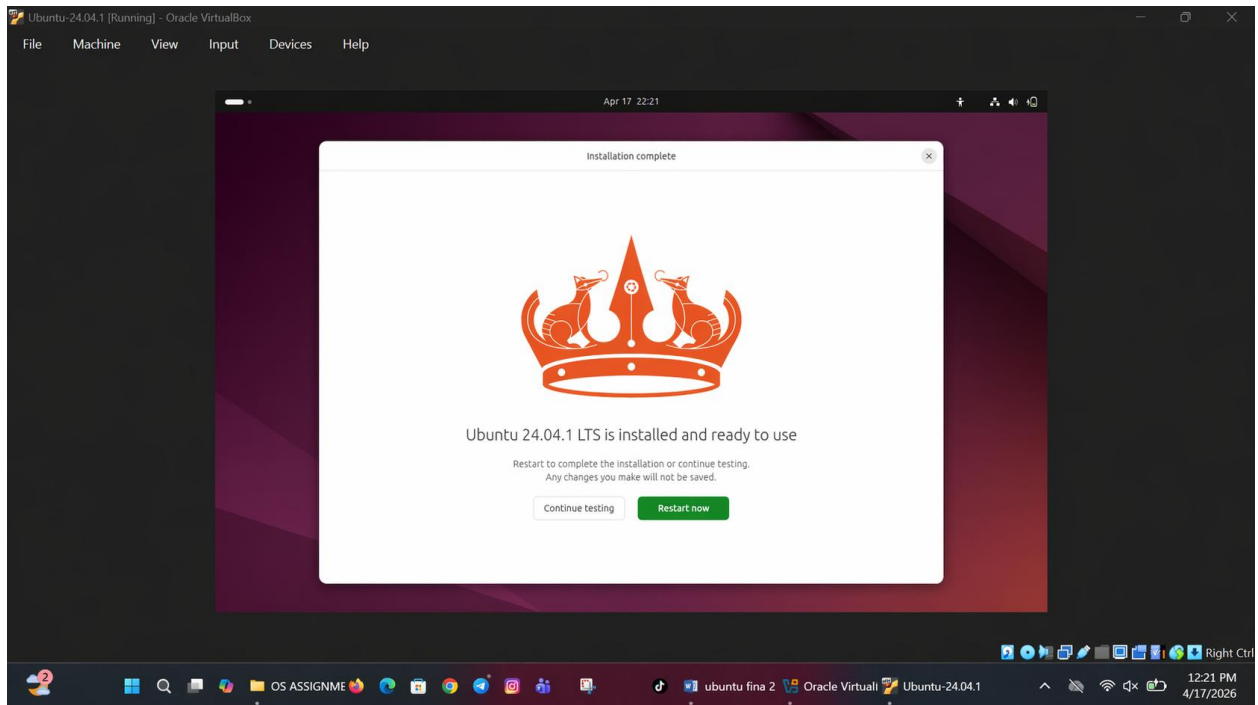
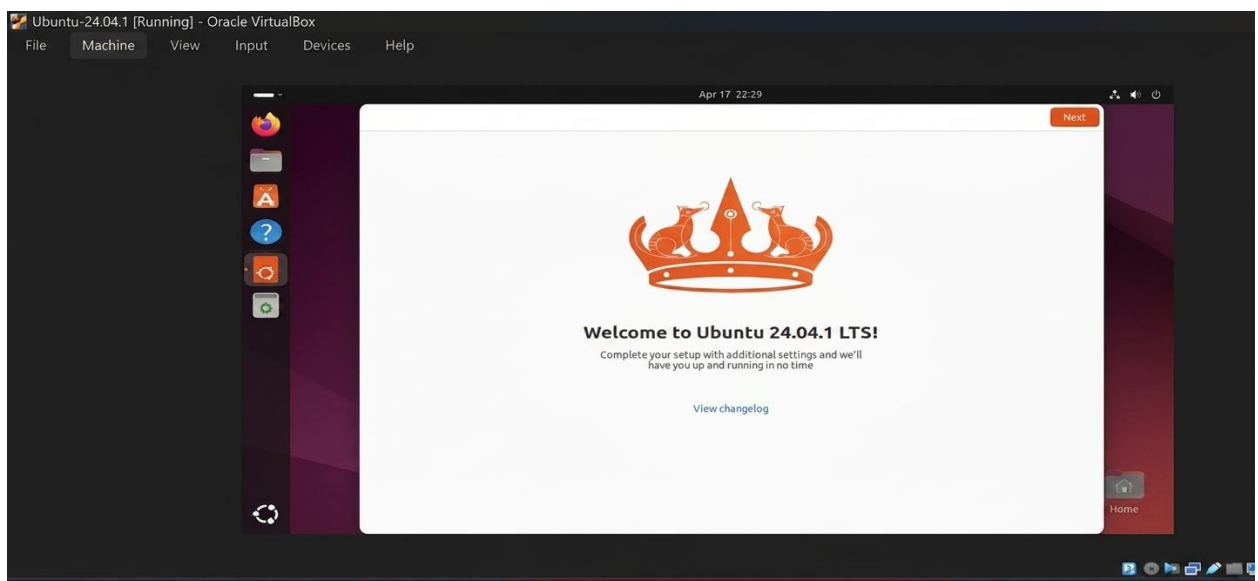


Figure 13: Restarting the system

Step 14: Login to Ubuntu

After restarting, enter your username and password that you created earlier. Once authenticated, you will successfully access the Ubuntu desktop environment.



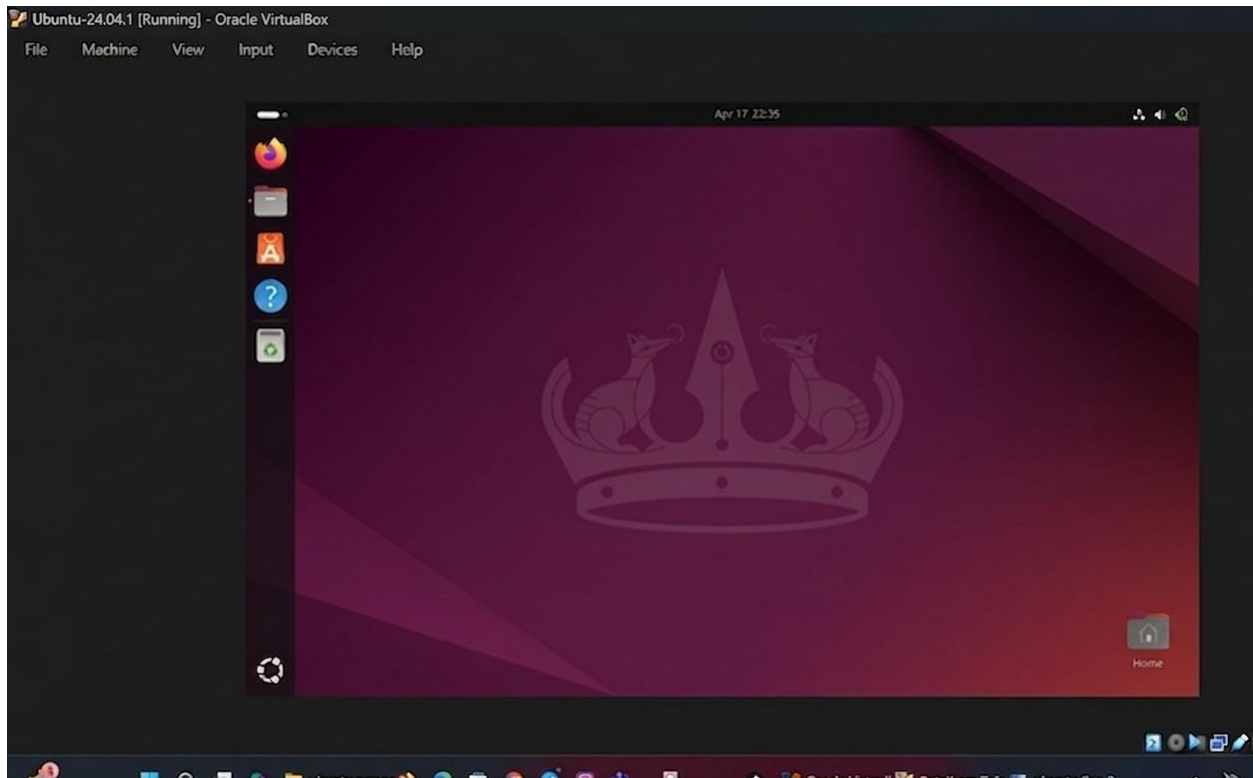


Figure 13 and 14: Ubuntu desktop login screen

4. Problems Faced

During the Ubuntu installation process inside the virtual machine, several common problems were experienced. One of the first issues was that the **virtual machine failed to start properly**, which prevented the installation from beginning. In some cases, a **black screen appeared after booting**, meaning the system loaded incorrectly or display settings were not configured well.

Another challenge was **slow system performance**, where the virtual machine responded slowly because of limited RAM or CPU resources. **Network connectivity problems** were also noticed, making it difficult to access the internet inside Ubuntu. In addition, there were **booting problems**, where the system could not load from the ISO file or restart correctly after installation.

5. Solutions

The above issues were solved using different troubleshooting methods. To fix startup problems, **virtualization was enabled in the BIOS settings**, allowing VirtualBox or VMware to run properly. For the black screen issue, the **display settings were adjusted**, such as changing graphics controller options or increasing video memory.

To improve slow performance, more resources were assigned by **increasing RAM and CPU allocation** for the virtual machine. Network issues were solved by **changing the network mode to NAT**, which allowed the guest operating system to access the internet through the host machine. Booting problems were fixed by **reattaching or reinstalling the Ubuntu ISO file**, ensuring the installation media was working correctly.

6. Filesystem Support

Ubuntu supports different types of file systems, and each one is designed for specific tasks and storage needs.

- **ext4 (default):** The most common Linux file system. It is fast, stable, reliable, and widely supported.
- **Btrfs:** A modern file system that offers advanced features such as snapshots and easier backup management.
- **NTFS:** Mainly used by Windows systems, but Ubuntu can read and write to it for compatibility.
- **FAT32 / exFAT:** Commonly used for USB drives, memory cards, and other external storage devices.
- **ZFS:** An advanced file system with strong data protection, storage pooling, and management features.

Why ext4 is Preferred

The **ext4** file system is preferred by many Ubuntu users because it provides excellent speed, strong stability, and reliable performance. It handles large files efficiently, reduces system crashes, and has wide support across Linux environments. Because of these advantages, ext4 is the default choice during most Ubuntu installations.

7. Advantages and Disadvantages

Advantages

- Easy to use and beginner-friendly
- Strong community and documentation
- Free and open-source
- Secure and stable
- Suitable for development and learning

Disadvantages

- Some proprietary software is not supported
- Requires basic Linux knowledge
- Hardware compatibility issues may occur in some cases

8. System Call: `getuid()`

What It Does

In Linux and Unix systems, every user account has a unique number called a **User ID (UID)**. The `getuid()` system call is used to return the UID of the user who is currently running a program.

In simple words, it helps the system know which user started the process.

For example:

- **Root user** usually has UID **0**
- Normal users often have UIDs such as **1000**, **1001**, and higher depending on the system setup

This allows the operating system to correctly identify the owner of a running process.

Importance of `getuid()`

The `getuid()` system call is useful for many reasons:

- **Identifies the current user** – It shows which user is running the program.
- **Permission control** – The system can decide whether the user has access to files or resources.
- **Improves security** – Programs can check if they are being run by an administrator or a normal user.
- **Useful for developers** – Many applications use it when user identity is required.

Example Program

```
#include <stdio.h>
#include <unistd.h>

int main() {
    printf("User ID: %d\n", getuid());
    return 0;
}
```

Explanation of the Program

The program includes the `unistd.h` header file because it contains the `getuid()` function. When the program runs, it calls `getuid()` and prints the current user's ID.

Example output:

```
User ID: 1000
```

This means the program is running under the account whose UID is **1000**.

```
}
```

9. Virtualization in Modern Operating Systems

What is Virtualization?

Virtualization is a modern technology that allows a computer to create a **virtual version of hardware, storage, or an entire operating system** inside the same physical machine. In simple words, it makes one computer behave like many separate computers. These virtual computers are called **Virtual Machines (VMs)**, and each one can run its own operating system independently.

For example, a user can install Windows as the main system and run Ubuntu Linux inside a virtual machine at the same time. This makes virtualization very useful in education, business, software development, and system administration.

Why is it Important?

Virtualization is important because it provides many benefits in modern computing environments. First, it creates a **safe environment for testing** new software, applications, or operating systems without affecting the main computer. If a problem happens inside the virtual machine, the host system remains safe.

Second, it helps to **reduce hardware cost** because one physical computer can run multiple virtual systems. Instead of buying many computers, organizations can use one powerful machine more efficiently.

Third, virtualization allows **multiple operating systems to run simultaneously**. A user can work on Windows, Linux, and other systems on the same computer without restarting.

Finally, it improves **security and isolation**. Each virtual machine works separately, so problems such as viruses or crashes in one machine usually do not affect the others.

How Does it Work?

Virtualization works through software called a **hypervisor**. A hypervisor is responsible for creating and managing virtual machines. It allocates hardware resources such as CPU, RAM, storage, and network access to each virtual system.

Popular hypervisors include **VirtualBox**, **VMware**, and **Hyper-V**. These tools allow a guest operating system to run independently on top of the host operating system. The host is the main system installed on the computer, while the guest is the virtual system running inside it.

For example, a student using Windows can install VirtualBox and create an Ubuntu virtual machine for learning Linux commands. This shows how virtualization makes computing flexible, efficient, and practical in modern operating systems.

Conclusion

This project provided a practical and meaningful introduction to operating systems through the installation and use of Ubuntu 24.04.1 in a virtual environment. By going through each step of the installation process, I gained a clearer understanding of how operating systems are set up and how they interact with hardware and software resources.

Working in a virtualized environment made it possible to experiment safely without affecting the main system, which is especially valuable for learning and testing. In addition, exploring the `getuid()` system call helped me understand how Linux identifies users and manages permissions, an essential concept in system programming and security.

Overall, this assignment strengthened both my theoretical knowledge and practical skills. It showed how important Linux-based systems are in modern computing and gave me confidence to continue exploring more advanced topics in operating systems and system programming.

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Future Outlook / Recommendation

Building on the knowledge gained from this project, there are several areas that can be explored further to deepen understanding and improve practical skills in operating systems and system programming.

First, it is important to continue practicing Linux commands and becoming more comfortable with the Ubuntu environment. This will help in performing tasks efficiently and understanding how the system works internally. In addition, studying more advanced system programming concepts, such as process management, memory handling, and file operations, will strengthen technical skills.

Another important area for future learning is exploring kernel-level operations to understand how the core of the operating system functions. Gaining experience in real-world development using Ubuntu can also be highly beneficial, as it allows the application of theoretical knowledge in practical situations.

Furthermore, learning about cybersecurity and system administration is highly recommended, since these fields are closely related to operating systems and are in high demand. Overall, continuous practice, exploration, and hands-on experience will be key to becoming proficient in this area.

References

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2. Oracle VM VirtualBox User Manual – Guide for creating and managing virtual machines.
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